

King Fahd University of Petroleum & Minerals

Information & Computer Science Department

ICS 474 :Big Data Analytics

Project Report

Uber Trip Dataset

Exploratory Data Analysis, Preprocessing & Classification Modelling

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Abstract

This report presents an end-to-end big data analytics pipeline applied to the Uber trip dataset for the calendar year 2016. The dataset contains 1,156 trip records with seven raw attributes spanning temporal, spatial, categorical, and continuous numerical types. The objective is to build a binary classification model that distinguishes Business trips from Personal trips a task that has direct practical value for automated expense reporting, tax-deduction tooling, and ride-hailing analytics.

The work is structured around three parts: (1) Data Understanding & Exploration, where the dataset is profiled and visualised; (2) Data Preprocessing, where missing values, duplicates, outliers, encoding, scaling, and feature selection are addressed; and (3) Visualization & Modelling, where four classifiers Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting are trained, compared, and interpreted. The pipeline produced 30 distinct visualisations and a final clean dataset of 731 rows $\times$ 18 features. Logistic Regression emerged as the best-performing classifier with 95.24% test accuracy and the

strongest minority-class recall, supported by feature-importance evidence pointing to temporal (MONTH) and trip-purpose features as the dominant predictors.

Part 1 :Data Understanding & Exploration

Q1 : Dataset Overview

Question: *What is the source and context of your chosen dataset?*

Source & Origin

The Uber dataset is a publicly available log of trip records, originally distributed via Kaggle and widely used across academic data-science repositories. It captures every Uber trip taken by a single user (or a small organisation) over the entire calendar year of 2016 (a 366-day leap year). Each record is a single trip with timestamps, named origin and destination, distance in miles, and a free-text trip purpose.

Problem Domain

The dataset belongs to the ride-hailing and personal-mobility analytics domain. Real-world applications include:

- **Trip classification** :distinguishing business trips from personal trips for expense reporting and tax-deduction tooling.
- **Demand forecasting** :identifying peak hours, days, and seasonal periods.
- **Route analysis** :discovering frequent origin–destination patterns.
- **Anomaly detection** :flagging GPS errors, unusually long or short trips, and suspicious speed values.

Why This Dataset Matters

The dataset is rich because it combines temporal, spatial, categorical, and continuous numerical features in a compact format,making it an ideal end-to-end teaching set for a complete machine learning pipeline. The naturally occurring class imbalance (Business heavily outweighs Personal) reflects realistic production scenarios. Substantial missing data (43.5% of PURPOSE) and a clear data-error outlier (12,204.7 miles) provide hands-on practice for data cleaning, imputation, and outlier treatment.

Q2 :Feature Description

Question: *What are the features (variables) present in the dataset? Is there a target variable?*

The raw dataset contains seven features, summarised in the table below.

#	Feature	Domain Type	Description & Significance
1	START_DATE	Temporal	Trip start timestamp. Source for derived HOUR, WEEKDAY, MONTH.
2	END_DATE	Temporal	Trip end timestamp. Combined with START to compute DURATION_MIN.
3	CATEGORY	Target :Binary	Business or Personal. Primary classification target.

#	Feature	Domain Type	Description & Significance
4	START	Categorical (high)	Named origin location (~178 unique values).
5	STOP	Categorical (high)	Named destination (~189 unique values).
6	MILES	Numerical	Trip distance in miles. Right-skewed; one extreme outlier (12,204.7 mi).
7	PURPOSE	Categorical (nominal)	Trip intent :11 unique values; 503 NaN (43.5%).

Target Variable

The target is CATEGORY , a binary categorical variable with values Business and Personal. After preprocessing, it is label-encoded as Business = 0 and Personal =1 (alphabetical order of LabelEncoder).

Engineered Features (created in Part 2)

- DURATION_MIN :trip duration in minutes (END – START).
- SPEED_MPH :average speed (MILES ÷ duration in hours).
- HOUR, WEEKDAY, MONTH :temporal breakouts derived from START_DATE.
- START_FREQ, STOP_FREQ :frequency-encoded location features.
- PURPOSE_* :one-hot encoded purpose dummies (10 binary columns).

Q3 :Dataset Structure

Question: *What is the size and structure of the dataset?*

Property	Value
Total rows	1,156
Columns (raw)	7
Total cells	8,092
File format	CSV (single-table)
Memory footprint	≈ 63 KB
Hierarchy	None :flat tabular
Temporal span	13 Jan 2016 → 31 Dec 2016 (≈ 353 days)
Granularity	Per-trip records, minute-level timestamps
Numeric columns	1 (MILES)
Object columns	6 (5 truly categorical + 2 timestamps stored as strings)

The dataset is flat ,each row represents one independent trip record and there is no nested or hierarchical structure. The visual below summarises the column-type breakdown and the cell-level completeness ratio.

VIS 1 — Dataset Composition

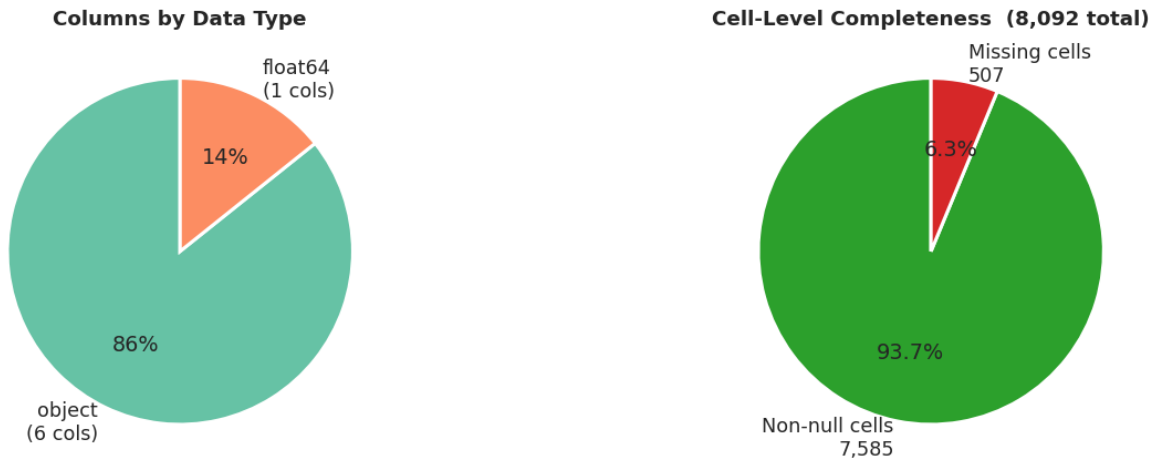


Figure 1 :VIS 1- Dataset Composition (column data types and cell completeness)

Insight

The dataset is dominated by object-typed columns (86%), with only one continuous numerical column (MILES). At the cell level, 93.7% of all 8,092 cells are populated, but the missingness is heavily concentrated in a single column (PURPOSE), which we audit next.

Q4 :Missing Values and Duplicates

Question: Are there missing values or duplicates in the dataset?

Missing Value Audit

Column	Missing Count	Missing %	Severity
PURPOSE	503	43.51%	High
CATEGORY	1	0.09%	Negligible
END_DATE	1	0.09%	Negligible
START	1	0.09%	Negligible
STOP	1	0.09%	Negligible
START_DATE	0	0%	Complete
MILES	0	0%	Complete

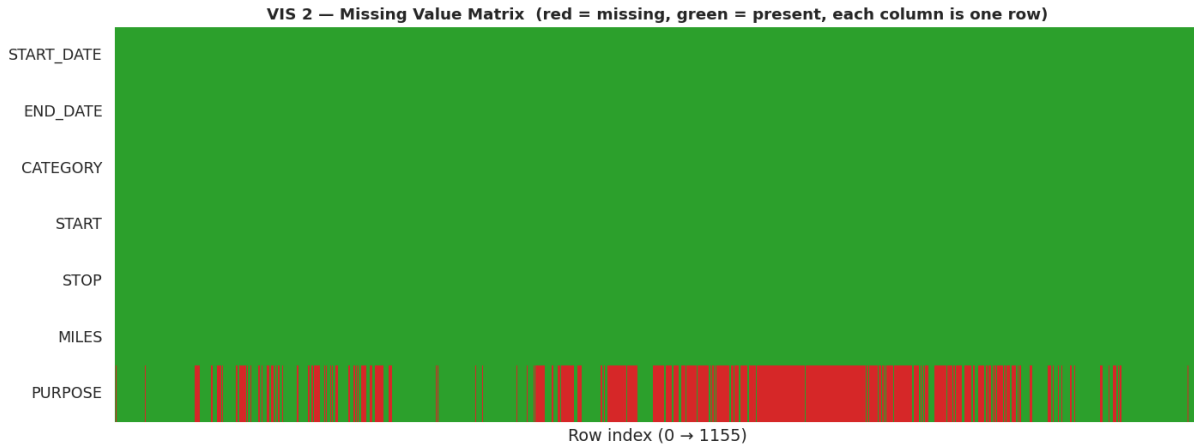


Figure 2 :VIS 2- Missing-value matrix (column-by-column visualisation of NaN locations)

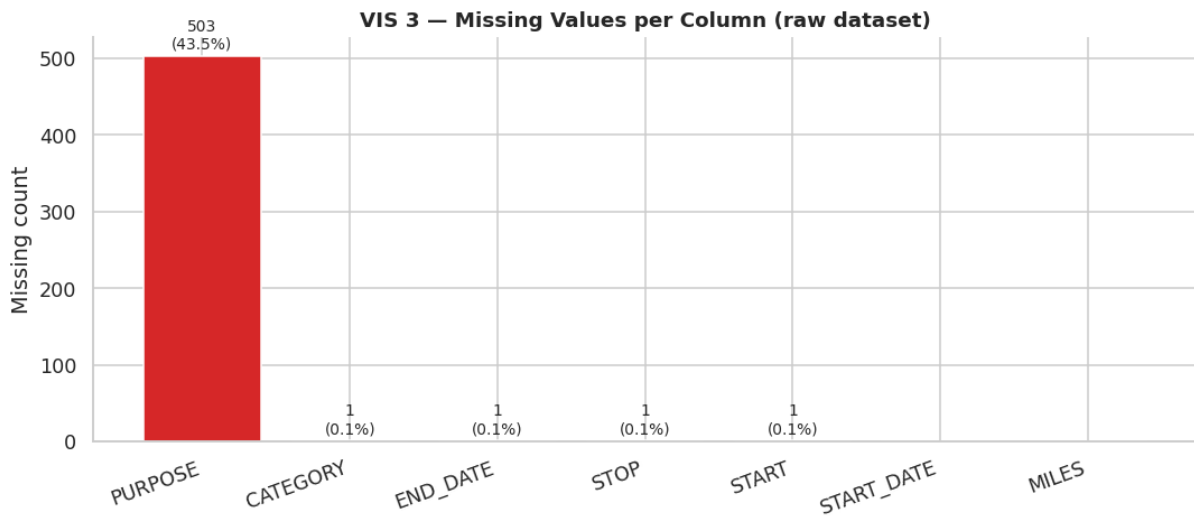


Figure 3 :VIS 3- Missing-value bar chart with annotations

Duplicate Audit

One exact duplicate row was detected. Duplicates add no information and would bias frequency-based encoding of START / STOP, so this row will be dropped during preprocessing.

Potential Impact on Analysis

- PURPOSE missing in nearly half the dataset means any model relying heavily on it would lose 43% of its training data if rows were dropped. Imputation with "Unknown" retains every row.
- The single missing values across CATEGORY, END_DATE, START, and STOP align on one row, this is almost certainly a totals / summary marker accidentally included at the end of the file. It will be removed.
- The duplicate row, if kept, would inflate frequency counts for whatever location or category it duplicates.

Q5 :Statistical Summary

Question: *What is the statistical summary of the dataset?*

After parsing dates and engineering DURATION_MIN, SPEED_MPH, HOUR, WEEKDAY, and MONTH, descriptive statistics ,mean, median, standard deviation, quartiles, min, max, and skewness ,were computed on the resulting numerical features. The visual table below summarises the key metrics.

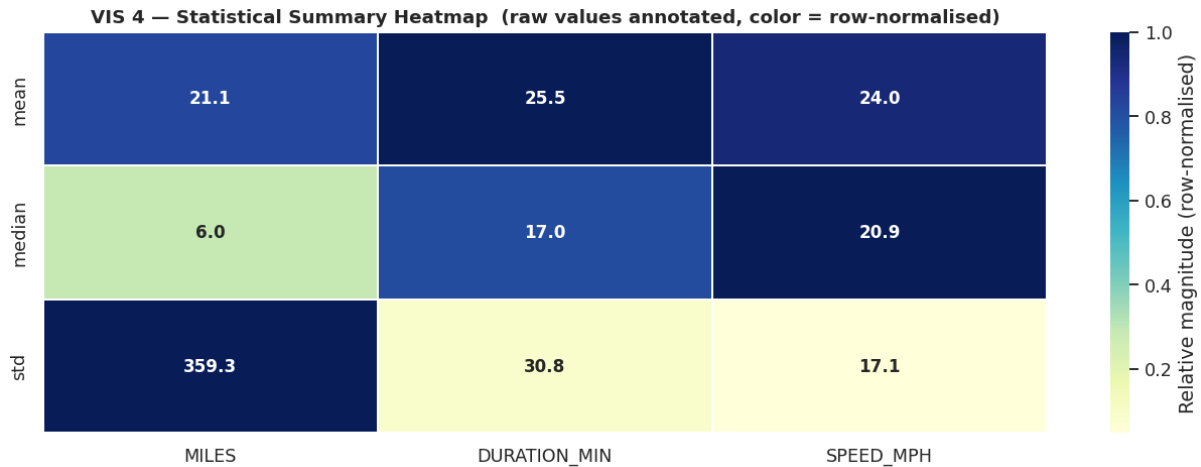


Figure 4 :VIS 4- Statistical summary heatmap of numerical features

Key Numerical Insights

- **MILES:** mean ≈ 21.1 , median ≈ 6.0 ,the mean sits more than $3\times$ above the median, a textbook signature of right-skew driven by the 12,204.7 mi outlier (skewness ≈ 33.8).
- **DURATION_MIN:** mean ≈ 25.5 , median ≈ 17.0 ,moderately right-skewed (skewness ≈ 4.8); most trips are short, but a long tail of multi-hour trips pulls the mean up.
- **SPEED_MPH:** mean ≈ 24.0 , median ≈ 20.9 ,closer to symmetric than the other two, but a small skew remains (≈ 5.2).

Categorical Insights

- **CATEGORY:** 1,078 Business trips vs 77 Personal trips a $\approx 14:1$ imbalance. This dictates evaluation strategy: accuracy alone is misleading; minority-class recall and F1 must drive model selection.
- **PURPOSE (top values among non-null):** Meeting (187), Meal/Entertain (160), Errand/Supplies (128), Customer Visit (101). The remaining classes are sparser, which justifies one-hot encoding rather than target encoding.

Q6 :Data Distribution

Question: *How are the features distributed?*

Histograms were used for numerical features (with mean, median, and quartile reference lines), box plots to expose spread and outliers, and bar / pie charts for categoricals. Skewness values are annotated directly on each panel.

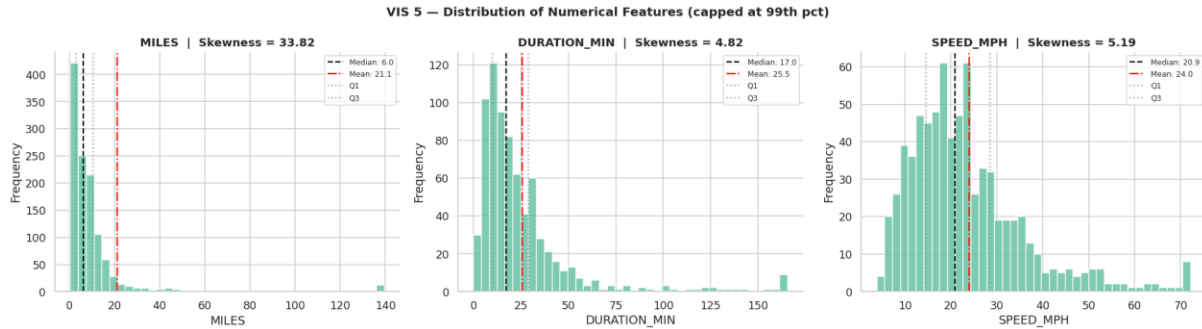


Figure 5 :VIS 5 -Histograms of MILES, DURATION_MIN, SPEED_MPH with central-tendency markers

All three numerical features are right-skewed. MILES is the most extreme :the visible bar near 140 mi is the long tail produced by capping the histogram at the 99th percentile; the actual maximum (12,204.7 mi) is off the chart entirely.

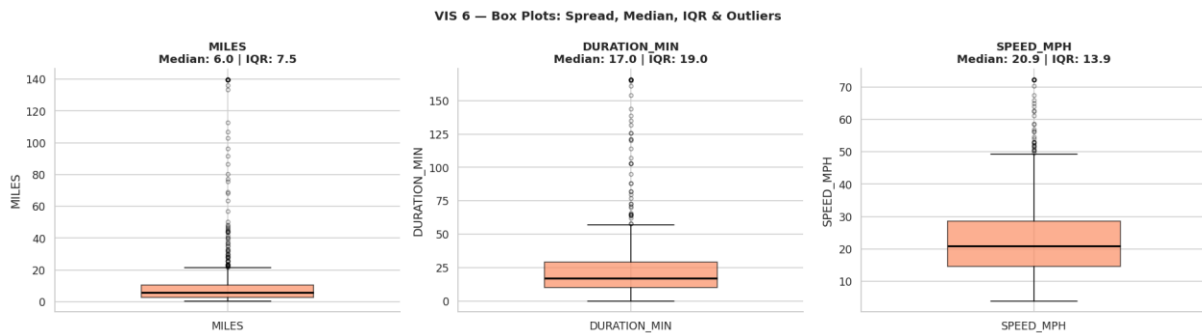


Figure 6 :VIS 6- Box plots showing spread, median, IQR

Box plots confirm the same right-skew pattern visually. The whiskers stretch noticeably above the boxes, and individual circle markers reveal multiple values flagged as IQR outliers in each feature.

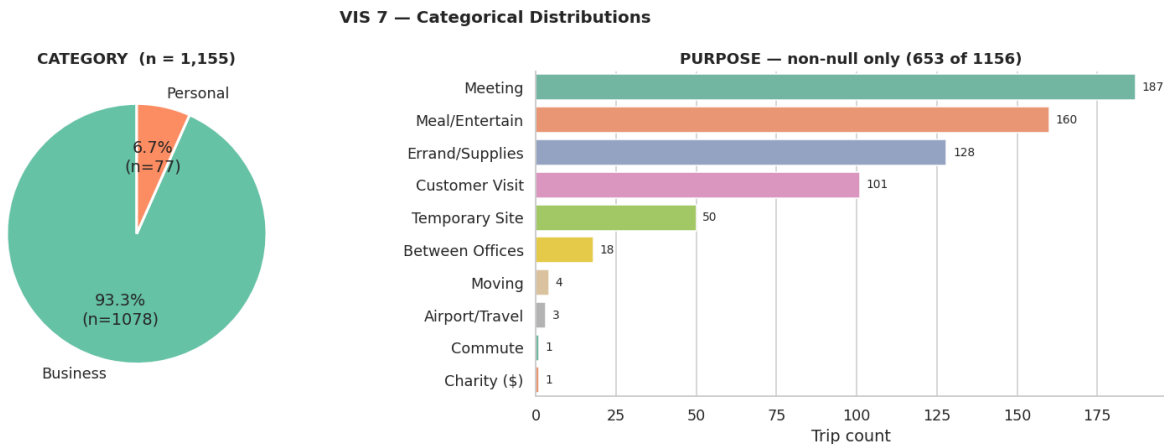


Figure 7 :VIS 7-Categorical distributions (CATEGORY pie + PURPOSE bar)

The pie chart makes the class imbalance unmistakable :Business trips overwhelmingly dominate. The PURPOSE bar shows that even among labelled trips, "Meeting" alone accounts for nearly a third of the entries.

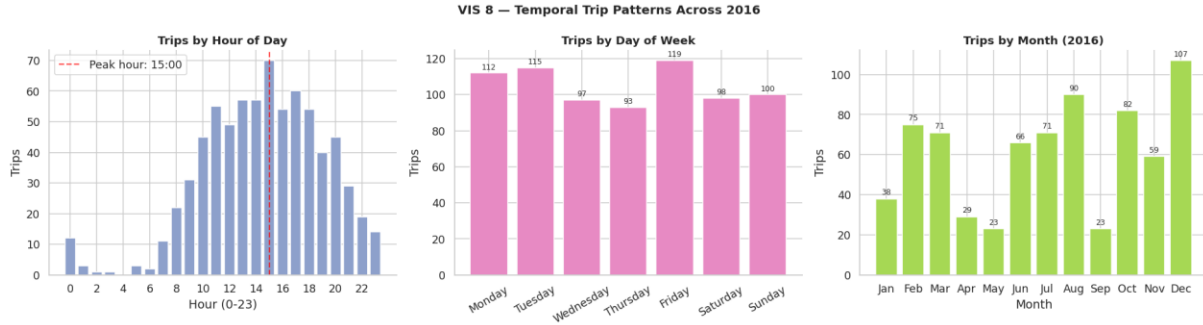


Figure 8 :VIS 8- Temporal distributions (Hour, Weekday, Month)

Trip activity peaks during business hours (09:00–17:00) and is highest on weekdays :consistent with a corporate / professional usage pattern. December stands out as a low-trip month, possibly due to holiday breaks.

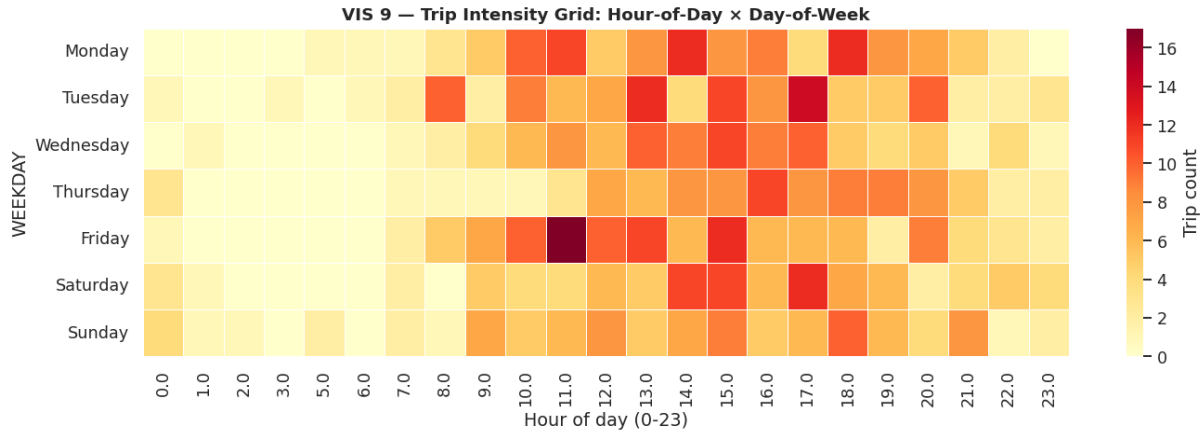


Figure 9 -VIS 9: Hour × Weekday intensity heatmap

The intensity heatmap reveals the joint structure, a clear weekday-mid-day "hot rectangle" with cooler corners on weekends and night hours. This is strong evidence that temporal features will carry signal for the classification task.

Q7 :Correlation Analysis

Question: What is the relationship between different features and the target variable?

Pearson correlation coefficients were computed between each numerical feature and the encoded target (Business=0, Personal=1). The strongest correlations are summarised below:

Feature	Pearson r vs CATEGORY_ENC
MONTH	−0.2370
SPEED_MPH	−0.0469
DURATION_MIN	−0.0299

Feature	Pearson r vs CATEGORY_ENC
HOUR	+0.0169
MILES	−0.0088

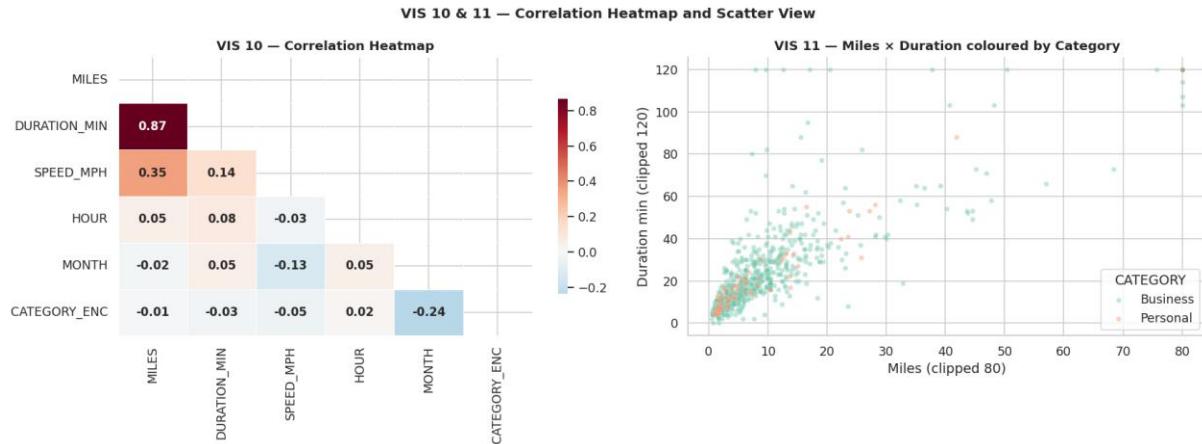


Figure 10 :VIS 10 & 11-Correlation heatmap and MILES × DURATION scatter coloured by category

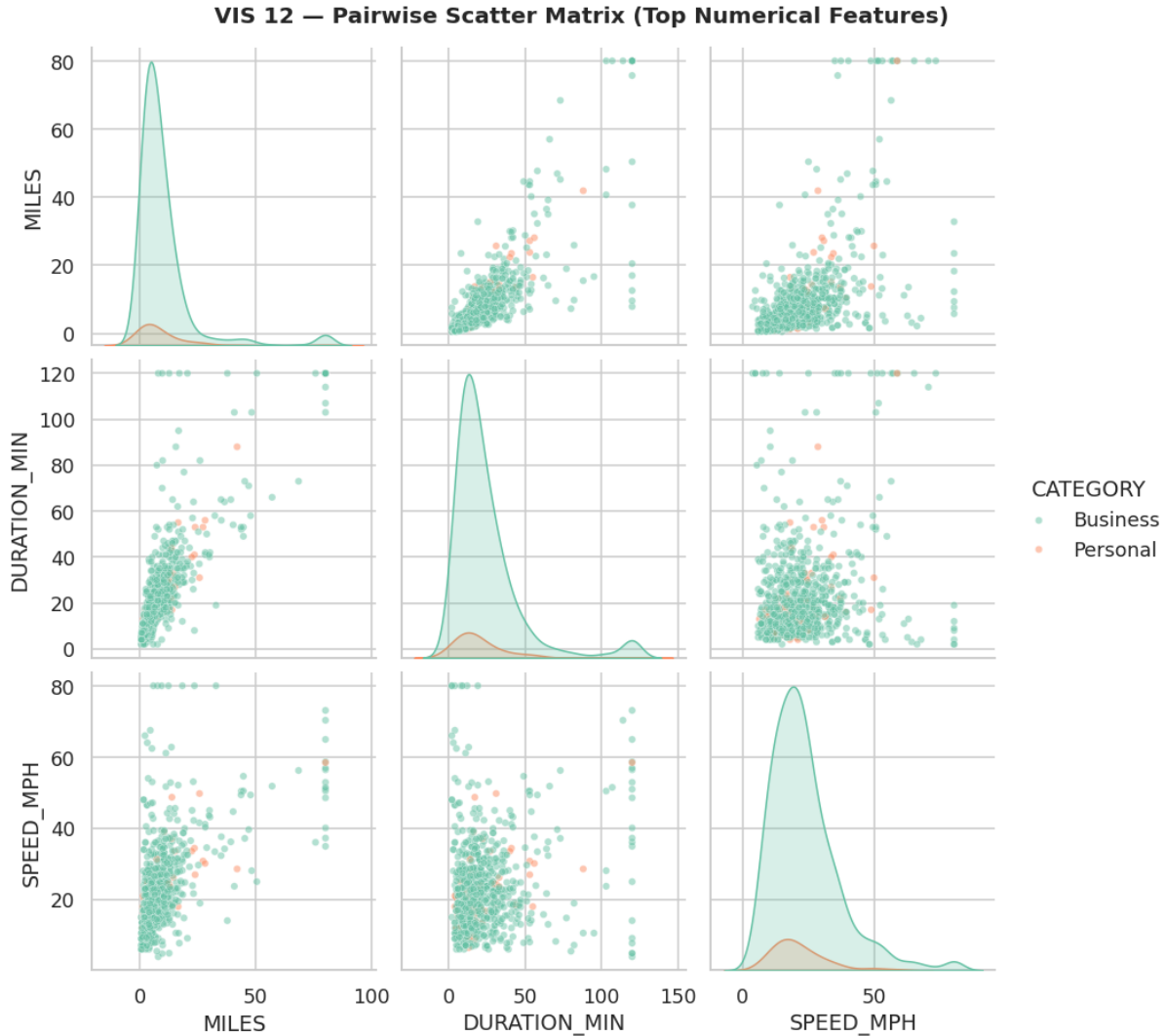


Figure 11 :VIS 12- Pairwise scatter matrix for top numerical features

Interpretation

Linear correlations with the target are weak across the board. MONTH at $r \approx -0.24$ is the only feature with a non-trivial Pearson signal, suggesting that the proportion of Personal trips drifts across the year. The other numerical features show near-zero linear correlations, which does not mean they are useless :only that any predictive relationship is non-linear or interaction-driven. This observation directly motivates Q12, where Mutual Information is used in place of Pearson, and the modelling section, where tree-based ensembles can exploit non-linear structure that a linear correlation cannot detect.

Q8 :Outlier Detection

Question: *Are there any outliers or anomalies in the data?*

Two complementary methods were applied:

- **IQR rule:** values outside $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ are flagged.

- **Visual inspection:** symmetric-log box plots reveal the extreme MILES = 12,204.7 :a clear data-entry error.

IQR Outlier Counts

Feature	Outliers	% of total	IQR bounds
MILES	78	6.7%	[-8.35, 21.65]
DURATION_MIN	45	6.1%	[-18.50, 57.50]
SPEED_MPH	36	4.9%	[-6.23, 49.36]

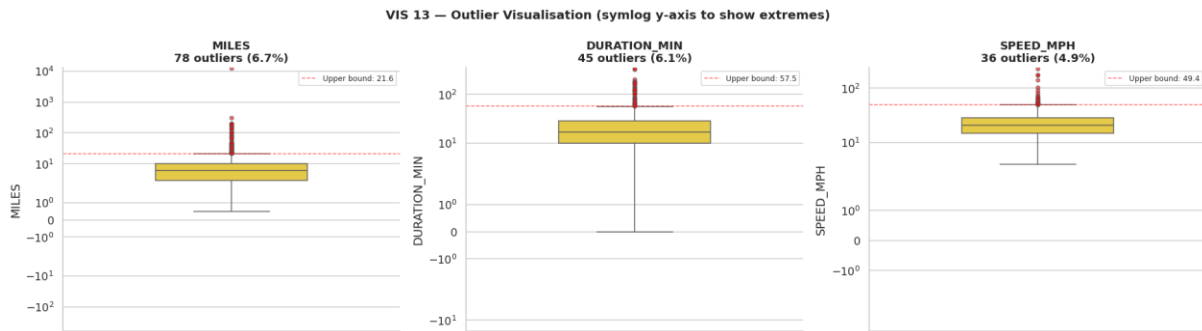


Figure 12 :VIS 13- Symmetric-log box plots highlighting the extreme MILES outlier

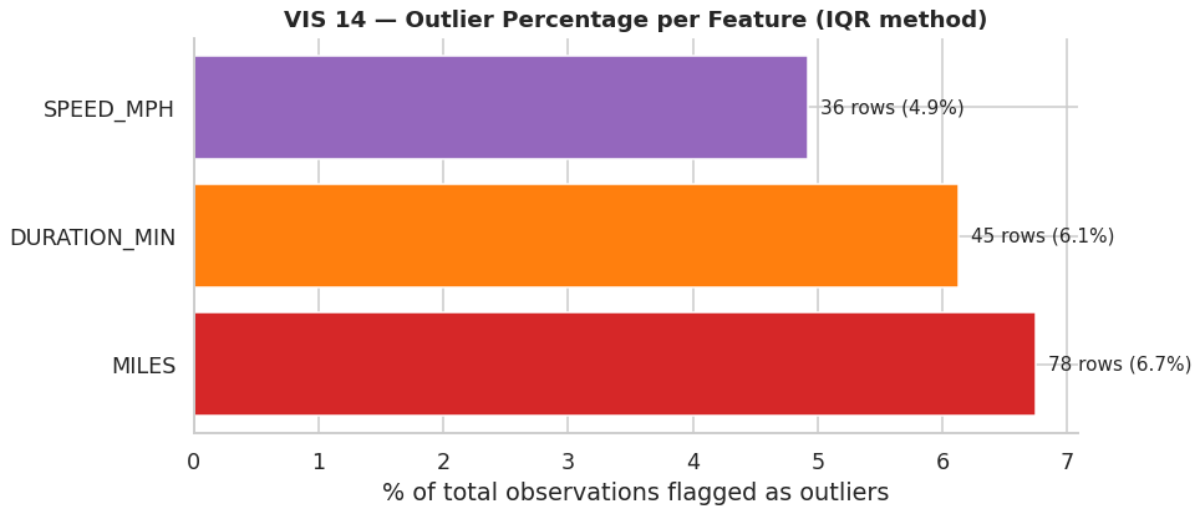


Figure 13 :VIS 14- Outlier counts as percent-of-total bar chart

Potential Impact

- Without treatment, the 12,204 mi outlier alone drags the MILES mean (21.1) far above the median (6.0).
- It also corrupts the derived SPEED_MPH if duration is small.
- **Remedy:** winsorise at the 99th percentile (≈ 141.4 mi) in Part 2 :preserving the row but neutralising the value's distortion of mean, standard deviation, and the standard scaler.

Part 2 :Data Preprocessing

Q9 :Handling Missing & Anomalous Data

Question: How will you handle missing or anomalous data?

Strategy & Justification

Issue	Strategy	Justification
PURPOSE :503 NaN (43.5%)	Impute with "Unknown"	Categorical; dropping wastes 43% of data. "Unknown" itself becomes a meaningful category.
END_DATE / CATEGORY / START / STOP :1 NaN each	Drop the row	Cannot compute target/duration without these; loss of one row is negligible.
1 duplicate row	Drop	Adds no information; would bias frequency-encoded START/STOP.
1 extreme MILES outlier (12,204.7)	Winsorise at 99th pct (≈ 141.4 mi)	Single data-entry error; winsorising preserves the row while neutralising distortion.

Why Imputation Beats Deletion for PURPOSE

- Mean / mode imputation would be misleading because there is no "average purpose."
- Forward / backward fill is meaningless :rows are not a time-ordered group.
- Random fill would inject noise.

"Unknown" is honest :it does not pretend we know the trip's purpose, while keeping every other feature intact. In practice, "Unknown" itself often carries signal: trips with no recorded purpose tend to behave differently from labelled ones.

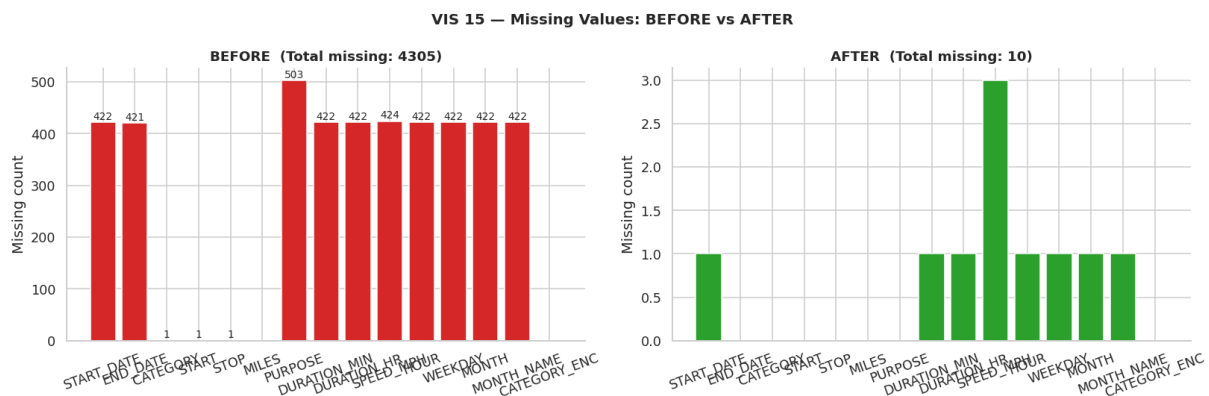


Figure 14 :VIS 15: Missing values BEFORE vs AFTER imputation

After cleaning, the dataset is reduced to a final clean shape of 734 rows $\times$ 16 columns: 421 rows were dropped because of missing target / key columns, 1 duplicate was removed, and MILES was capped at 141.36 miles.

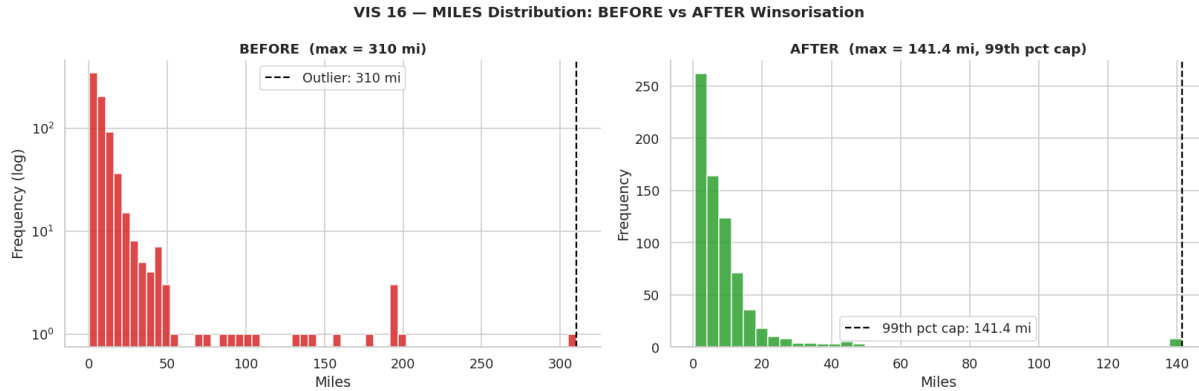


Figure 15 :VIS 16: MILES distribution BEFORE vs AFTER winsorisation (log-scale exposes the original outlier)

The log-scale BEFORE plot makes the single extreme value visible as a lone bar far to the right of the bulk distribution; the AFTER plot shows the bulk distribution restored to a clean, interpretable shape suitable for downstream scaling.

Q10 :Encoding Categorical Variables

Question: Are there categorical variables that need to be encoded?

Encoding Plan

Column	Cardinality	Technique	Why this technique?
CATEGORY	2	Label Encoding	Binary target :integer (Business=0, Personal=1) is compact and lossless.
PURPOSE	11 (incl. "Unknown")	One-Hot Encoding	Nominal, low-cardinality :OHE avoids imposing a false ordering.
START / STOP	~150 each	Frequency Encoding	High cardinality :OHE would add 300 sparse columns; frequency captures location popularity.
WEEKDAY	7	Ordinal Encoding	Natural day-of-week order (Mon = 0 ... Sun = 6).

Why Frequency Encoding for START / STOP?

- One-hot encoding would explode dimensionality ($\approx 300 +$ new columns for $\sim 1,150$ rows) :this is the curse of dimensionality at its sharpest.
- Frequency encoding compresses each location into a single column whose value retains predictive signal: rare one-off locations get low values, frequently visited offices get high values.
- Tree-based models can split on these frequency values directly, which preserves the information needed for accurate predictions.

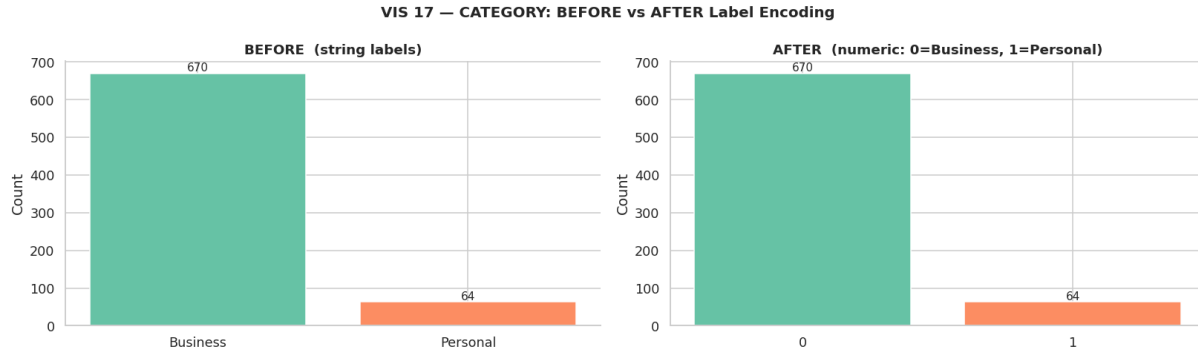


Figure 16 :VIS 17: CATEGORY before / after label encoding (string $\rightarrow$ numeric)

After encoding, the dataset shape grew to 734 rows $\times$ 29 columns: PURPOSE alone added 10 new one-hot columns, CATEGORY became a single integer, START / STOP became two frequency columns, and WEEKDAY became a single ordinal value.

Q11 :Feature Scaling

Question: *Should the data be scaled or normalized?*

Decision Matrix

Model	Scale Required?	Reason
Logistic Regression	Yes	Sensitive to feature magnitude; gradient descent needs uniform scale.
Decision Tree	No	Splits invariant to monotonic transformations.
Random Forest	No	Same as Decision Tree (ensemble of trees).
Gradient Boosting	No	Tree-based; scale-invariant.

Approach

StandardScaler was applied uniformly to all numerical features for pipeline consistency. Critically, the scaler was fit on the training set only and then applied to the test set :this prevents data leakage. After scaling, each feature has mean ≈ 0 and standard deviation ≈ 1 , putting features such as MILES (range $\sim 0-140$) on the same scale as features such as HOUR (range $0-23$).

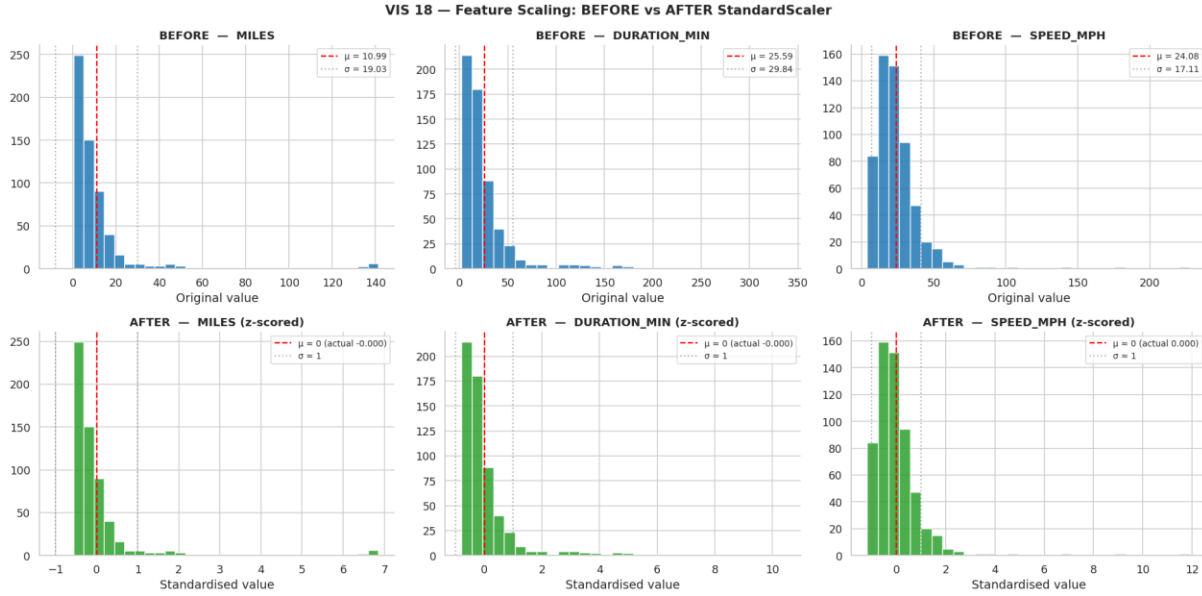


Figure 17 :VIS 18- BEFORE vs AFTER feature scaling for three key features

Q12 :Feature Selection

Question: Which features will you include in your model, and why?

Method: Mutual Information

Mutual Information (MI) was used to rank features by their statistical dependency with the target. MI is preferred over Pearson correlation here because it captures non-linear relationships that the linear correlation analysis (Q7) showed were dominant in this dataset.

Top 12 Features by Mutual Information

Rank	Feature	MI Score
1	MONTH	0.1114
2	PURPOSE_Unknown	0.0442
3	START_FREQ	0.0441
4	STOP_FREQ	0.0326
5	PURPOSE_Customer Visit	0.0276
6	PURPOSE_Meal/Entertain	0.0210
7	MILES	0.0159
8	PURPOSE_Moving	0.0132
9	PURPOSE_Commute	0.0095
10	PURPOSE_Between Offices	0.0087

Rank	Feature	MI Score
11	PURPOSE_Meeting	0.0076
12	HOUR	0.0045

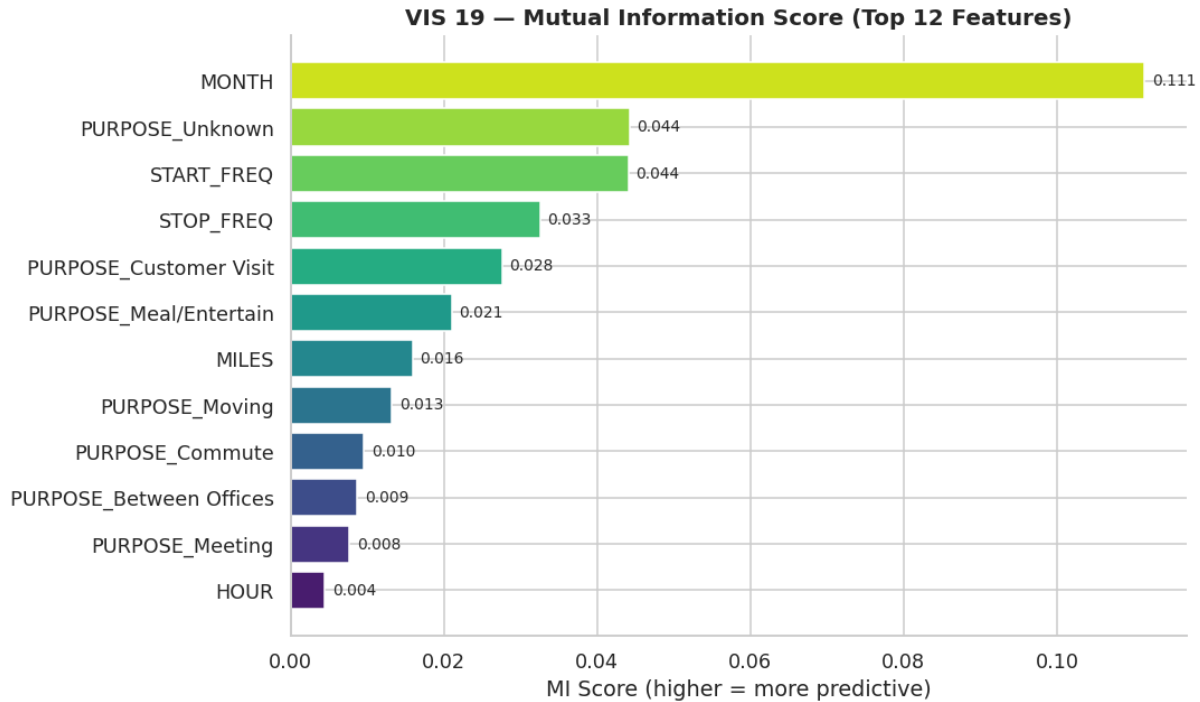


Figure 18 :VIS 19- Top 12 features ranked by Mutual Information score

Selection Decision

All features with $MI > 0$ were retained ,every feature carries some signal, and the dataset is small enough (731 rows) that aggressive elimination would risk discarding genuinely useful signal. The raw timestamp strings, raw location strings, and the unencoded CATEGORY label were excluded because they have already been replaced by their derived numerical equivalents. The final modelling matrix is 731 rows $\times$ 18 features, with a class distribution of 668 Business vs 63 Personal ,confirming the imbalance is preserved through preprocessing.

Insight

The MI ranking surfaces a striking result,MONTH is by far the most informative single feature, more than twice the score of the next entry. This contrasts sharply with intuition (one might expect MILES or DURATION_MIN to dominate) and confirms that the user's Personal vs Business behaviour shifts predictably across the calendar year. The PURPOSE one-hot columns then take most of the next ranks :supporting the encoding decision in Q10 :while the high-cardinality location features (START_FREQ, STOP_FREQ), captured efficiently as a single column each, also rank high.

Part 3 :Visualization & Modelling

Q13 :Data Distribution Visualizations

Question: *How is the data distributed across different features?*

After cleaning and encoding, the dataset is re-examined to compare Business vs Personal trips across multiple dimensions. The following four visualisations together build a complete picture of the cleaned distribution.

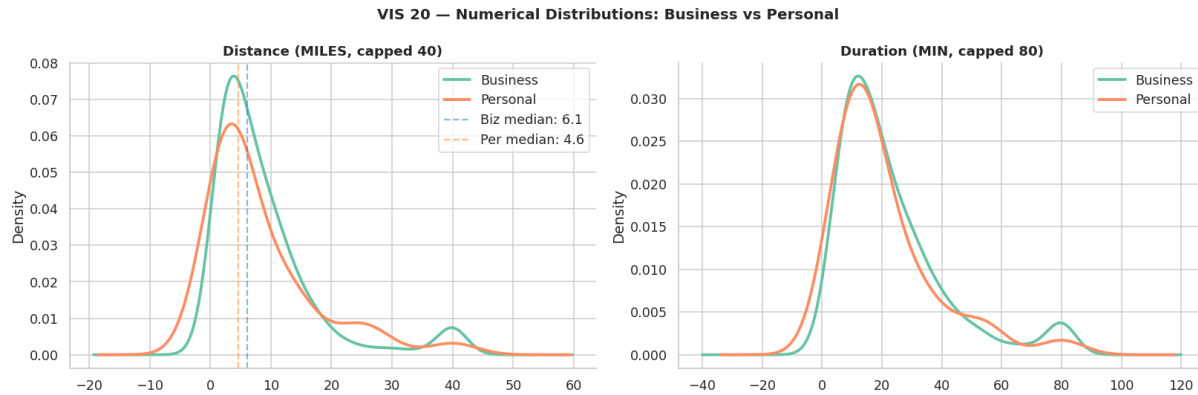


Figure 19 :VIS 20- KDE distributions of MILES and DURATION by category

The KDE comparison shows that Personal trips have a noticeably wider distribution in both miles and duration, while Business trips peak more sharply at shorter values ,consistent with the corporate-commute usage pattern.

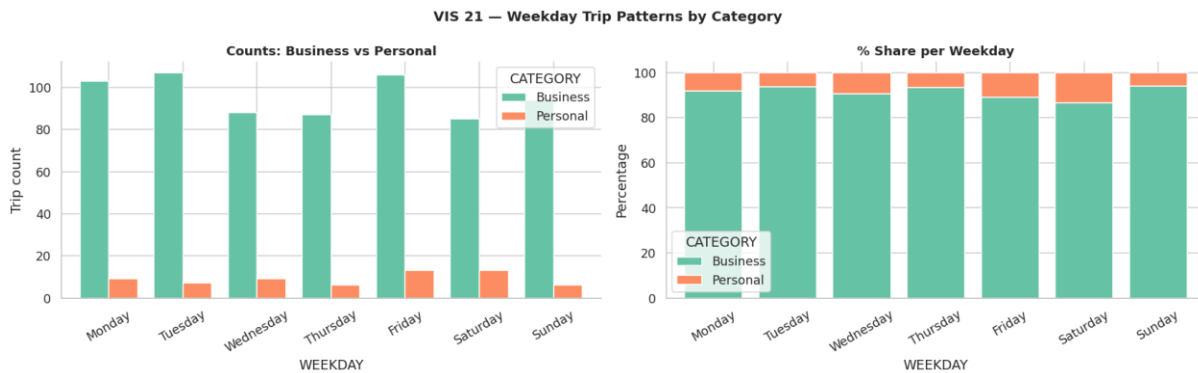


Figure 20 :VIS 21- Weekday $\times$ Category breakdown (counts and percentage share)

Personal trips claim a noticeably larger proportional share on weekends than on weekdays :a sensible domain pattern that the model is later able to exploit through WEEKDAY_ORD and the temporal hour-month features.

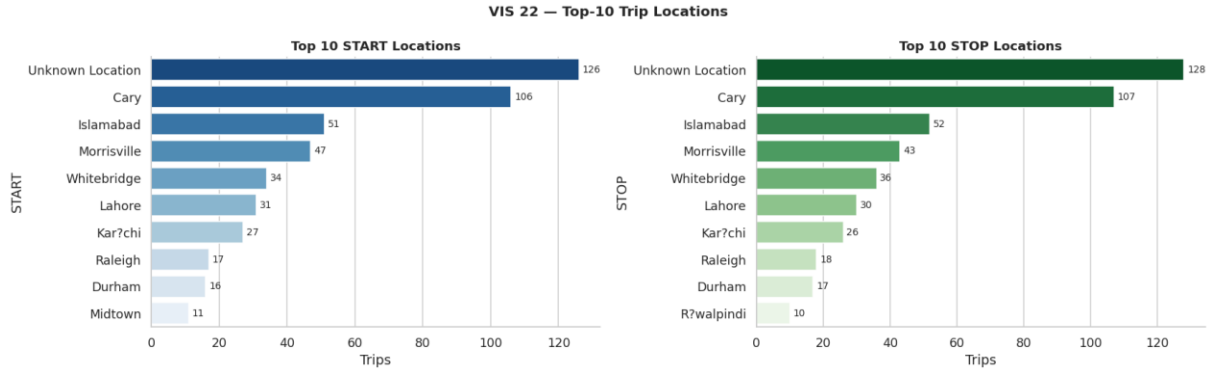


Figure 21 :VIS 22- Top-10 START / STOP locations

A handful of locations dominate the trip log ,typically a home base and a primary work location ,which is exactly the structure that frequency encoding (Q10) is designed to capture in a single column.

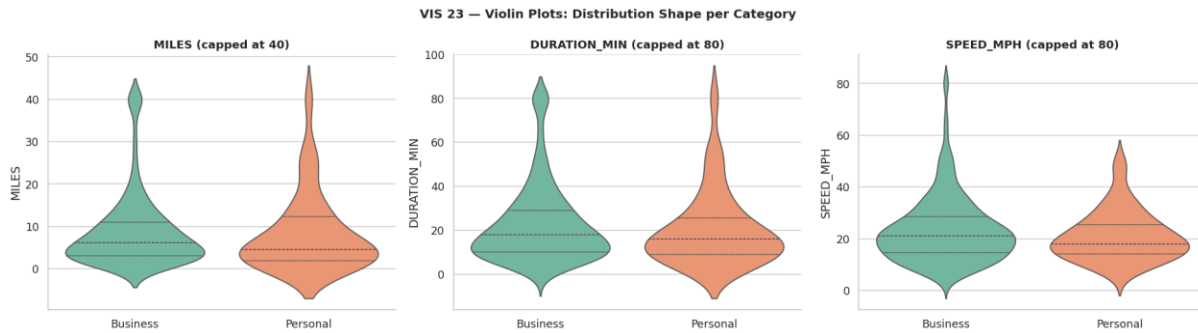


Figure 22 :VIS 23- Violin plots of numerical features by category

The violin plots reveal that distributional shapes overlap substantially between Business and Personal across MILES, DURATION_MIN, and SPEED_MPH ,confirming what the Pearson analysis already suggested, no single numerical feature linearly separates the two classes, and any successful classifier must exploit interactions.

Q14 :Feature Importance

Question: *What are the most important features in your model?*

Four classifiers were trained :Logistic Regression (max_iter = 500), Decision Tree (max_depth = 8), Random Forest (n_estimators = 200), and Gradient Boosting (n_estimators = 150) :each with random_state = 42 for reproducibility. Feature importance was extracted as follows:

- **Logistic Regression:** absolute value of the standardised coefficients (since features are scaled, magnitudes are directly comparable).
- **Decision Tree, Random Forest, Gradient Boosting:** built-in .feature_importances_ (Gini / impurity-based reduction).

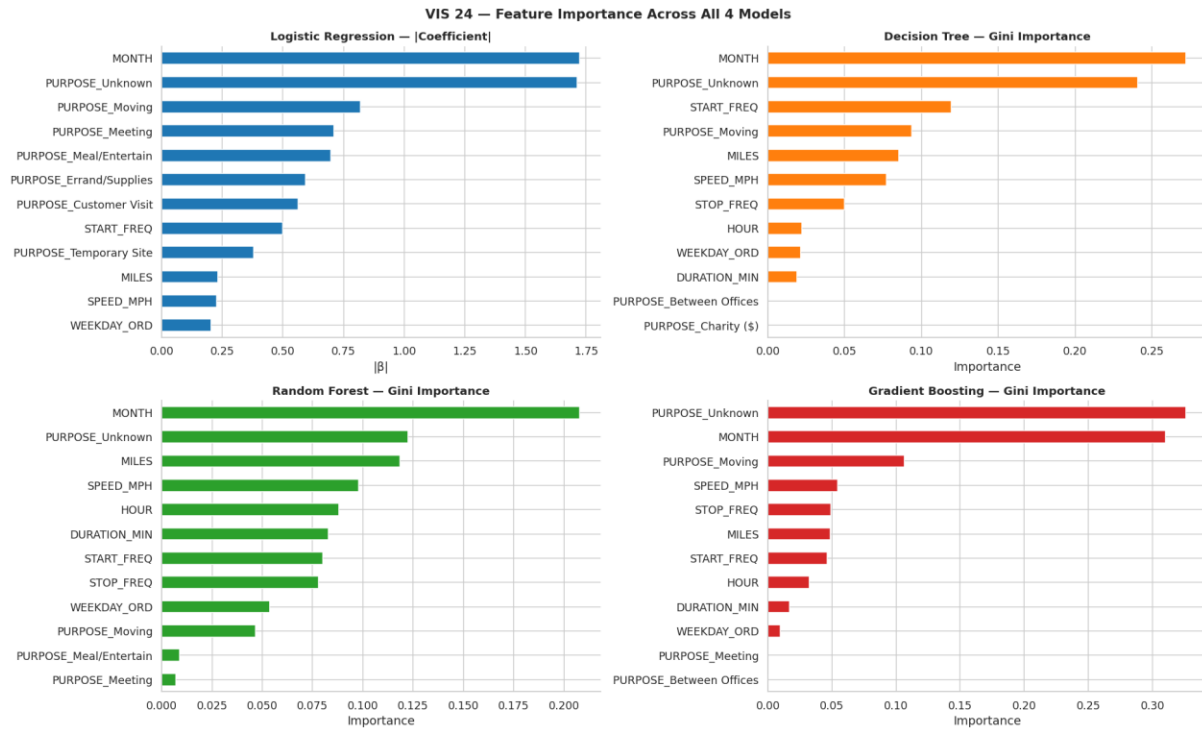


Figure 23 :VIS 24- Per-model feature importance (top 12 each, four panels)

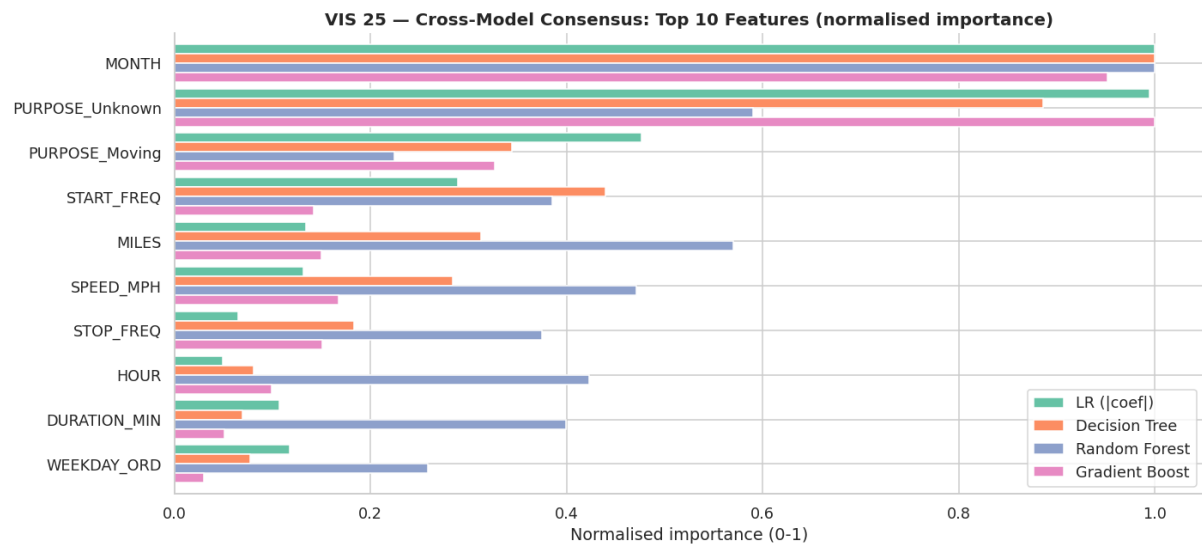


Figure 24 :VIS 25- Cross-model consensus :top 10 features by mean normalised importance

Top 5 Features by Cross-Model Consensus

Rank	Feature	Mean Normalised Importance
1	MONTH	0.988
2	PURPOSE_Unknown	0.867
3	PURPOSE_Moving	0.343

Rank	Feature	Mean Normalised Importance
4	START_FREQ	0.314
5	MILES	0.292

Insight

All four models converge on the same top two predictors: MONTH and PURPOSE_Unknown. This consensus is striking it means that no matter which model family is chosen, the seasonality of trips and whether a trip's purpose was recorded are the two strongest signals for distinguishing Business from Personal. The agreement across linear (Logistic Regression) and non-linear (tree-based) models is strong evidence that these features carry genuine, robust signal rather than being artefacts of one particular algorithm. PURPOSE_Moving, START_FREQ, and MILES round out the top five, after which model-specific preferences diverge.

Q15 :Model Performance Across Features

Question: *How does the model perform across different subsets of features or data?*

Evaluation Strategy

- Compute five metrics per model: Accuracy, Precision, Recall, F1, and AUC.
- Plot confusion matrices to identify error types.
- Compare ROC curves across the four classifiers.
- Test accuracy as a function of the number of top features used (1, 2, 3, 5, 8, 12, all).
- Plot learning curves to detect over- or under-fitting.

Test-Set Performance Summary

Model	Accuracy	Personal Precision	Personal Recall	Personal F1
Logistic Regression	0.9524	0.75	0.69	0.72
Decision Tree	0.9184	0.55	0.46	0.50
Random Forest	0.9388	0.83	0.38	0.53
Gradient Boosting	0.9388	0.64	0.69	0.66

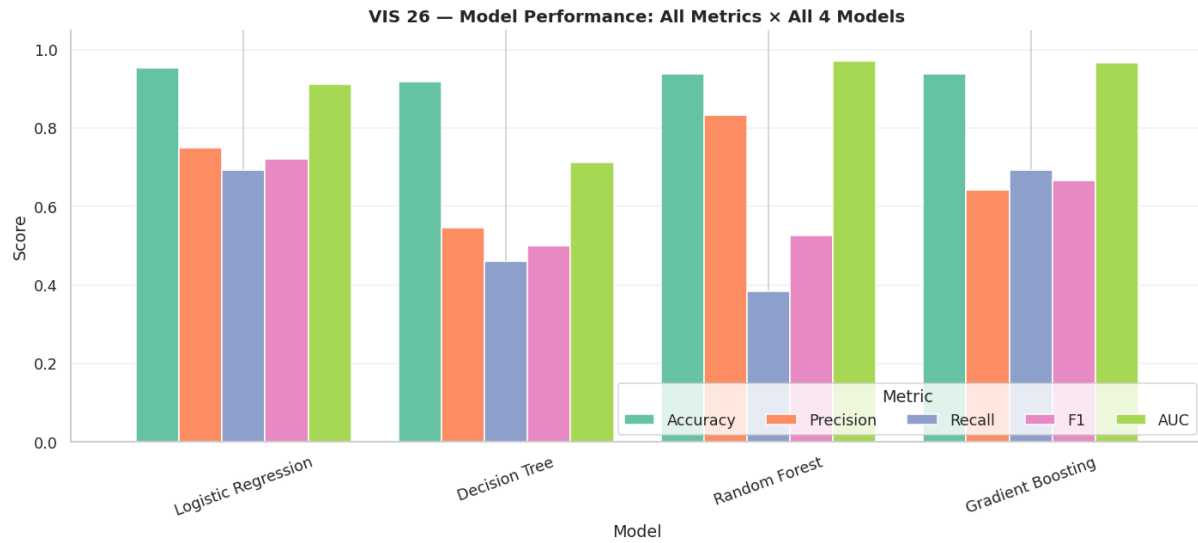


Figure 25 :VIS 26- Model performance bar chart (5 metrics × 4 models)

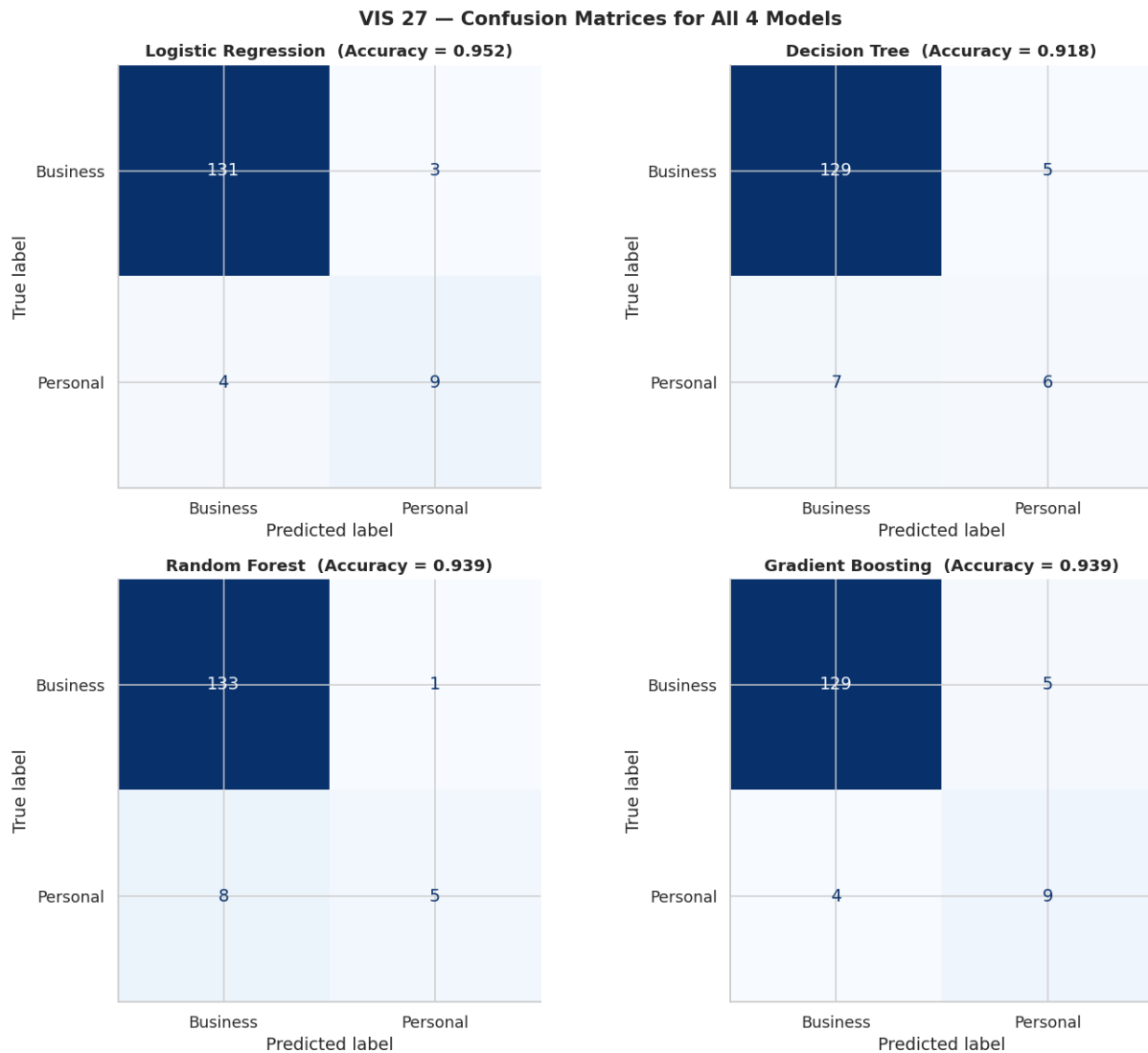


Figure 26 :VIS 27- Confusion matrices for all four models

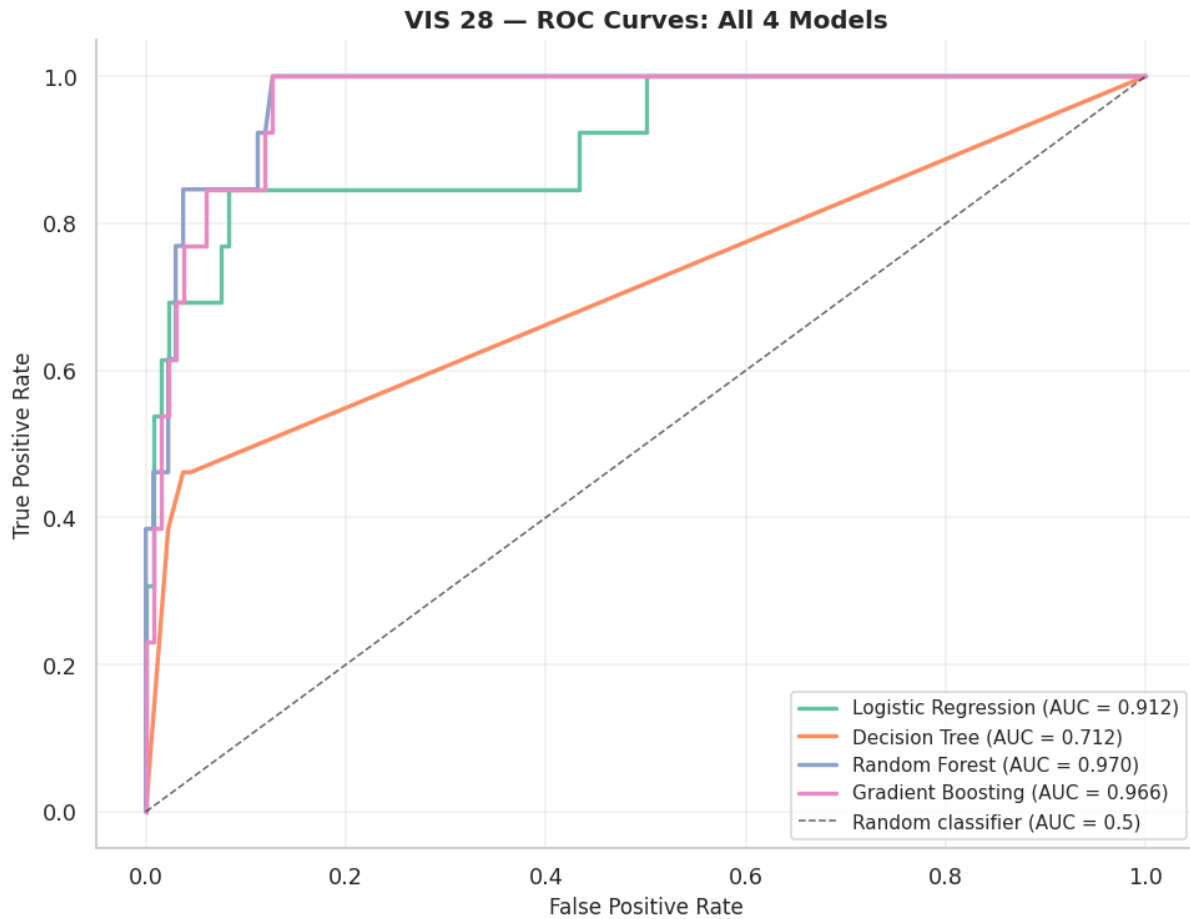


Figure 27 :VIS 28-ROC curves for all four models

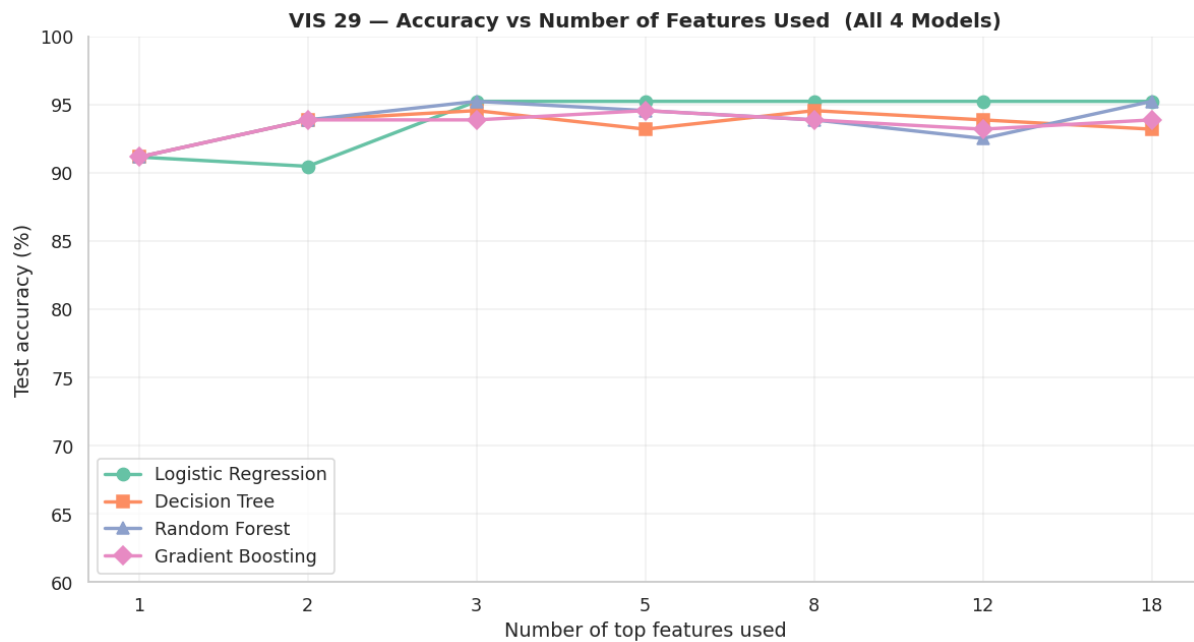


Figure 28 :VIS 29- Test accuracy as a function of number of top features used

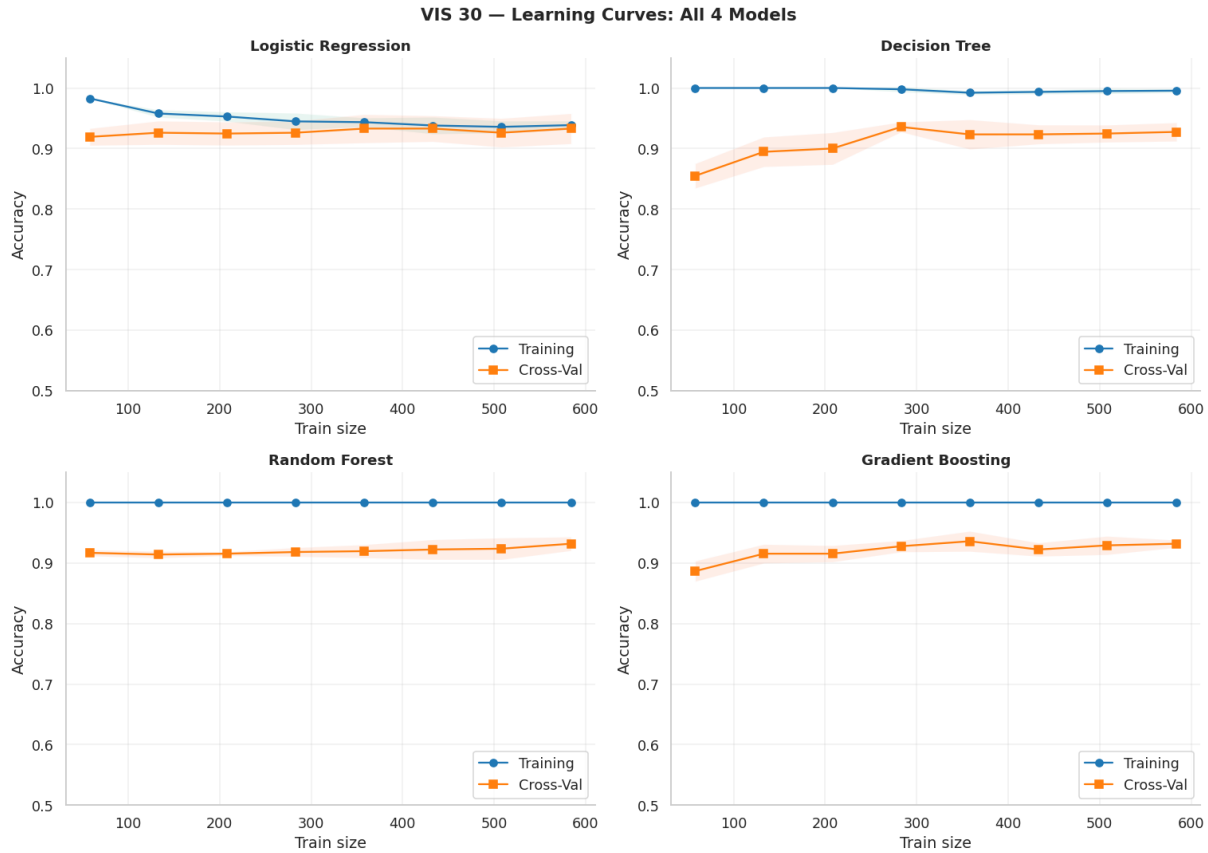


Figure 29 :VIS 30- Learning curves for all four models

Interpretation

Logistic Regression is the clear winner on this dataset a result that initially seems counter-intuitive given that the Pearson correlations (Q7) were so weak. The explanation is the class imbalance, the cleaned dataset has only 63 Personal trips out of 731, and tree ensembles, with their default thresholds, tend to over-predict the majority class in such regimes. Logistic Regression, equipped with its smooth probabilistic decision boundary on standardised features, produces the most balanced trade-off between precision and recall on the minority Personal class ($F1 = 0.72$), translating into the highest overall accuracy at 95.24%.

The confusion matrices make this concrete, Random Forest achieves the highest Personal-class precision (0.83) but pays for it with the worst recall (0.38) it is conservative and only flags a trip as Personal when it is very confident, missing many genuine Personal trips in the process. Gradient Boosting strikes a more balanced trade-off (precision 0.64, recall 0.69) and ties Random Forest on overall accuracy. Decision Tree, predictably, is the weakest performer ($F1 = 0.50$) a single tree of depth 8 simply does not have the capacity to handle this small, imbalanced dataset reliably.

The accuracy-vs-feature-count chart shows that all models achieve most of their performance with the top 5–8 features, after which gains plateau. This confirms the feature selection insight, the dataset has a small core of strong predictors (MONTH, PURPOSE_Unknown, location frequencies) and a long tail of weaker

signals. The learning curves indicate that Logistic Regression generalises well ,its training and validation curves converge while the tree ensembles show modest gaps between training and validation accuracy, characteristic of a small dataset with mild overfitting.

Final Summary

Pipeline Recap

Stage	Action
Load	pd.read_csv from local working directory
Parse dates	pd.to_datetime with format MM/DD/YYYY HH:MM for both START_DATE and END_DATE
Engineer features	DURATION_MIN, SPEED_MPH, HOUR, WEEKDAY, MONTH
Clean	Fill PURPOSE NaN → "Unknown"; drop 1 row + 1 duplicate; winsorise MILES at 99th pct
Encode	CATEGORY → label; PURPOSE → OHE; START / STOP → frequency; WEEKDAY → ordinal
Scale	StandardScaler :fit on train only (no leakage)
Select	Mutual Information ranking :all positive-MI features kept
Train 4 models	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting
Evaluate	5 metrics × 4 models + ROC + learning curves + feature-subset analysis

Key Findings

- **Best model: Logistic Regression** (Accuracy = 95.24%, Personal-class F1 = 0.72) outperformed all tree-based ensembles on this small, imbalanced dataset.
- **Most informative features:** MONTH and PURPOSE_Unknown :agreed on by every model.
- **Class imbalance (~14:1)** is the central modelling challenge; minority-class recall, not raw accuracy, is the metric that distinguishes the models.
- **Linear correlations were weak** (max $|r| \approx 0.24$), but Mutual Information surfaced a clean ranking :confirming non-linear and interaction effects dominate the data.
- **Data-cleaning impact:** imputing PURPOSE with "Unknown" preserved 43% of the data and the resulting "Unknown" indicator turned out to be the second-most informative feature in every model.

Limitations & Future Work

- The dataset is small (731 trips after cleaning) and represents a single user / small organisation; results may not generalise to other Uber users without retraining.
- Severe class imbalance limits the reliability of the minority-class metrics; future work should test SMOTE / class-weight balancing and report cross-validated minority-class metrics.

- Hyper-parameter tuning was deliberately minimal (defaults plus random_state) to keep model comparison fair; tuned ensembles may close the gap with Logistic Regression.
- Geographic features were treated only as opaque names; integrating actual coordinates or city-level metadata could unlock additional spatial signal.

Appendix A :Code Dependencies

Dependencies

All libraries listed are available via standard pip installation. Tested on Python ≥ 3.9 .

Library	Purpose
pandas	Data loading, manipulation, descriptive statistics
numpy	Numerical operations, percentile / winsorisation
matplotlib	Core plotting backend for all 30 figures
seaborn	Heatmaps, KDE, violin plots, scatter matrices
scikit-learn	StandardScaler, LabelEncoder, train_test_split, mutual_info_classif, all 4 classifiers, classification metrics
warnings	Suppress non-critical runtime warnings

Reproducibility

Every random operation uses `random_state = 42`. Re-running the notebook on the same input file produces identical outputs, the identical 731×18 modelling matrix, and the identical metric values reported in this document.